15: Model Selection

Stat 230 - Spring 2026

```
library(ggformula)
library(bayesrules) # data library
library(broom)
```

> **i** Note
>
> This activity goes with the Day 15 notes. Record your results in the table in your notes.

Overview

Today you will build a model to answer this question: **How are weekends and holidays associated with ridership, after accounting for the weather?** Run the code below to select the columns we will use and fit five candidate models.

```
bikes <- bikes[, c("date", "season", "year", "month", "weekend", "holiday",
                   "temp_actual", "temp_feel", "humidity", "windspeed",
"rides")]

bikes_A <- lm(rides ~ temp_actual + windspeed + weekend, data = bikes)
bikes_B <- lm(rides ~ temp_actual + windspeed + weekend + holiday, data =
bikes)
bikes_C <- lm(rides ~ temp_actual + windspeed + weekend + holiday + humidity,
data = bikes)
bikes_D <- lm(rides ~ temp_actual + windspeed + weekend + holiday + humidity +
temp_feel, data = bikes)
bikes_E <- lm(rides ~ temp_feel + humidity + weekend + holiday, data = bikes)
```

Part 1: Look before you fit

Q1. Make a correlation matrix of the weather variables. Which pair of predictors could cause problems if we include both in the same model? What problem?

```
bikes |>
  dplyr::select(temp_actual, temp_feel, windspeed, humidity) |>
  cor()
```

```
            temp_actual  temp_feel  windspeed   humidity
temp_actual   1.0000000  0.9823728 -0.1055362  0.2658495
temp_feel     0.9823728  1.0000000 -0.1462235  0.2707869
windspeed    -0.1055362 -0.1462235  1.0000000 -0.3244458
humidity      0.2658495  0.2707869 -0.3244458  1.0000000
```

`temp_actual` and `temp_feel` are almost perfectly correlated ($r = 0.98$). Including both would cause **multicollinearity**: they carry nearly the same information, so the model can't separate their effects, and the standard errors of their coefficients will be inflated. The other pairs of weather variables are only weakly or moderately correlated (all $|r| \leq 0.32$).

Part 2: Nested models and the F test

Q2. Which of these pairs of models are nested? For the pairs that are nested, which is the full model?

- A and B
- B and C
- C and D
- C and E
- A and C

- A and B: nested; B is full (adds `holiday`)
- B and C: nested; C is full (adds `humidity`)
- C and D: nested; D is full (adds `temp_feel`)
- C and E: **not nested**. E uses `temp_feel` but not `temp_actual` or `windspeed`, so neither model is a special case of the other.
- A and C: nested; C is full (adds `holiday` and `humidity`, so $d = 2$)

The `anova()` function takes the **reduced** model first and the **full** model second.

Q3. Fill in the blanks to run the extra sums of squares F test comparing models A and B, and then models B and C. What do you conclude each time?

```
anova(bikes_A, bikes_B)
```

```
Analysis of Variance Table

Model 1: rides ~ temp_actual + windspeed + weekend
Model 2: rides ~ temp_actual + windspeed + weekend + holiday
  Res.Df       RSS Df Sum of Sq      F   Pr(>F)
1    496 755019445
2    495 743486506  1  11532939 7.6784 0.005799 **
```

```
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(bikes_B, bikes_C)
```

```
Analysis of Variance Table

Model 1: rides ~ temp_actual + windspeed + weekend + holiday
Model 2: rides ~ temp_actual + windspeed + weekend + holiday + humidity
  Res.Df       RSS Df Sum of Sq     F    Pr(>F)
1    495 743486506
2    494 633868635  1 109617871 85.43 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

A vs. B: $H_0 : \beta_{holiday} = 0$ vs. $H_A : \beta_{holiday} \neq 0$. $F = 7.68$ on 1 and 495 df, p-value 0.006. We reject H_0: holidays are associated with ridership, adjusting for temperature, windspeed, and weekend.

B vs. C: $H_0 : \beta_{humidity} = 0$. $F = 85.4$ on 1 and 494 df, p-value < 0.001. We reject H_0: adding `humidity` removes a large amount of squared error (SSE falls from 743,486,506 to 633,868,635).

Q4. Now compare C to D (adding `temp_feel`). What does the F test suggest?

```
anova(bikes_C, bikes_D)
```

```
Analysis of Variance Table

Model 1: rides ~ temp_actual + windspeed + weekend + holiday + humidity
Model 2: rides ~ temp_actual + windspeed + weekend + holiday + humidity +
    temp_feel
  Res.Df       RSS Df Sum of Sq      F  Pr(>F)
1    494 633868635
2    493 629673075  1   4195560 3.2849 0.07053 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

$F = 3.28$ on 1 and 493 df, p-value 0.071. At the 5% level, we do **not** reject H_0: there is only weak evidence that `temp_feel` helps once `temp_actual` is already in the model. (This is borderline, which is a hint that something odd is going on. See Q5.)

Q5. Compare the `temp_actual` row of `summary(bikes_C)` to the `temp_actual` row of `summary(bikes_D)`. What changed, and why? How does this connect to your answer to Q1?

```
summary(bikes_C)
```

```

Call:
lm(formula = rides ~ temp_actual + windspeed + weekend + holiday +
    humidity, data = bikes)

Residuals:
    Min      1Q  Median      3Q     Max
-4721.7  -873.6   -81.6   803.7  2591.8

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   708.873    409.049   1.733  0.08372 .
temp_actual    94.878      5.206  18.226  < 2e-16 ***
windspeed     -66.721      9.702  -6.877 1.86e-11 ***
weekendTRUE  -777.129    113.562  -6.843 2.30e-11 ***
holidayyes  -1092.743    334.844  -3.263  0.00118 **
humidity      -33.435      3.617  -9.243  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1133 on 494 degrees of freedom
Multiple R-squared:  0.4877,    Adjusted R-squared:  0.4825
F-statistic: 94.06 on 5 and 494 DF,  p-value: < 2.2e-16
```

```
summary(bikes_D)
```

```

Call:
lm(formula = rides ~ temp_actual + windspeed + weekend + holiday +
    humidity + temp_feel, data = bikes)

Residuals:
    Min      1Q  Median      3Q     Max
-4751.5  -860.2   -53.0   793.5  2523.4

Coefficients:
             Estimate Std. Error t value Pr(>|t|)
(Intercept)   610.737    411.682   1.484  0.13858
temp_actual    46.308     27.297   1.696  0.09044 .
windspeed     -62.714      9.929  -6.316 5.99e-10 ***
weekendTRUE  -785.614    113.397  -6.928 1.34e-11 ***
```

```
holidayyes  -1069.943    334.309  -3.200  0.00146 **
humidity      -33.317      3.610  -9.230  < 2e-16 ***
temp_feel      45.069     24.866   1.812  0.07053 .
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 1130 on 493 degrees of freedom
Multiple R-squared:  0.4911,    Adjusted R-squared:  0.4849
F-statistic:  79.3 on 6 and 493 DF,  p-value: < 2.2e-16
```

In model C, `temp_actual` is hugely significant (estimate 94.9, SE 5.2, p < 0.001). In model D, once `temp_feel` is added, the standard error jumps to 27.3 (about 5.2 times bigger), the estimate drops to 46.3, and the p-value rises to 0.09. `temp_feel` also looks "insignificant" (p = 0.071), even though temperature clearly matters.

This is multicollinearity, which we predicted in Q1 ($r = 0.98$; the VIFs are about 30). The two variables carry nearly the same information, so the model can't tell which one deserves the credit. The overall fit barely changed, but the individual coefficients and t-tests became unreliable. The solution is to pick *one* temperature variable.

Part 3: Comparing models that aren't nested

Q6. Use `summary(...)$adj.r.squared`, `AIC()`, and `BIC()` to fill in the table in your notes for all five models.

```
models <- list(A = bikes_A, B = bikes_B, C = bikes_C, D = bikes_D, E =
bikes_E)

tibble::tibble(Model = names(models),
               k = sapply(models, function(m) length(coef(m))),
               adj_R2 = sapply(models, function(m) summary(m)$adj.r.squared),
               AIC = sapply(models, AIC),
               BIC = sapply(models, BIC)) |>
  knitr::kable(digits = c(0, 0, 3, 1, 1))
```

Model	k	adj_R2	AIC	BIC
A	4	0.386	8542.8	8563.8
B	5	0.394	8537.1	8562.4
C	6	0.483	8459.3	8488.8
D	7	0.485	8458.0	8491.7
E	5	0.445	8493.0	8518.3

This is the answer for the table in the notes. Smaller AIC and BIC are better; larger adjusted R^2 is better.

Q7. Models C and E are not nested. Which model do adjusted R^2, AIC, and BIC prefer? Why can't we use an F test to compare them?

Model C is preferred by all three: it has the higher adjusted R^2 (0.483 vs. 0.445) and the lower AIC (8459.3 vs. 8493) and BIC (8488.8 vs. 8518.3). We can't use an F test because the models are not nested: the F test compares a model to a special case of itself, but neither C nor E is a special case of the other.

Q8. Do the criteria agree on adding `temp_feel` to model C (comparing C and D)? What does each one say? Which model would you choose?

They do not agree. Adjusted R^2 barely changes (0.483 for C vs. 0.485 for D) and very slightly favors D. AIC is only 1.3 lower for D (a small difference), while BIC is 2.9 *higher* for D, so BIC prefers C. The F test (Q4) was borderline (p = 0.071).

When the tools disagree and there is multicollinearity, choose the **simpler** model: **model C**.

Part 4: Interpreting your model

Q9. Which of the five models would you use to answer our question? Justify your choice using what you found above.

Model C (`temp_actual + windspeed + weekend + holiday + humidity`):

- It contains our two variables of interest (`weekend` and `holiday`), and adjusts for the weather.
- Adding `holiday` (A to B) and `humidity` (B to C) both improved the fit significantly (F tests).
- Adding `temp_feel` (D) doesn't help much and causes multicollinearity.
- It beats the non-nested alternative E under every criterion.

(A reasonable alternative answer is B or C, as long as the choice is justified. Models A and E are weaker choices: A leaves out `holiday` and E has a worse fit.)

Q10. Use `confint()` on your chosen model. Interpret the coefficients for `weekend` and `holiday` (and their confidence intervals) in context. Why do you think the interval for `holiday` is so wide? (*Hint:* `table(bikes$holiday)`)

```
confint(bikes_C)
```

```
                  2.5 %     97.5 %
(Intercept)   -94.81684 1512.56232
temp_actual    84.65029  105.10609
windspeed     -85.78330  -47.65782
weekendTRUE -1000.25311 -554.00513
holidayyes  -1750.63757 -434.84846
humidity      -40.54199  -26.32735
```

```
table(bikes$holiday)
```

```
 no yes
488  12
```

Weekend: Holding temperature, windspeed, humidity, and holiday constant, weekend days have about 777 fewer rides than weekdays. We are 95% confident the true difference is between 554 and 1000 fewer rides.

Holiday: Holding the other variables constant, holidays have about 1093 fewer rides than non-holidays. We are 95% confident the true difference is between 435 and 1751 fewer rides.

The holiday interval is much wider because there are only 12 holidays among the 500 days in the data, so we have very little information about the holiday effect.

Q11. (*Extension*) Let's see what happens when we add some other candidate variables to model C. Compare C to each of the models below. Are these nested comparisons? What do the F test, adjusted R^2, AIC, and BIC say? Does adding them change your answer to our question?

```
bikes_F <- lm(rides ~ temp_actual + windspeed + weekend + holiday + humidity +
season,
              data = bikes)
bikes_Y <- lm(rides ~ temp_actual + windspeed + weekend + holiday + humidity +
year,
              data = bikes)
```

(*Note:* `season` has four levels, so adding it adds three coefficients, i.e., $d = 3$.)

```
anova(bikes_C, bikes_F)
```

```
Analysis of Variance Table

Model 1: rides ~ temp_actual + windspeed + weekend + holiday + humidity
Model 2: rides ~ temp_actual + windspeed + weekend + holiday + humidity +
```

```
    season
  Res.Df       RSS Df Sum of Sq      F    Pr(>F)
1    494 633868635
2    491 542812749  3  91055886 27.455 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
anova(bikes_C, bikes_Y)
```

```
Analysis of Variance Table

Model 1: rides ~ temp_actual + windspeed + weekend + holiday + humidity
Model 2: rides ~ temp_actual + windspeed + weekend + holiday + humidity +
    year
  Res.Df       RSS Df Sum of Sq      F    Pr(>F)
1    494 633868635
2    493 313266680  1 320601955 504.54 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
tibble::tibble(Model = c("C", "C + season", "C + year"),
               adj_R2 = sapply(list(bikes_C, bikes_F, bikes_Y), function(m)
summary(m)$adj.r.squared),
               AIC = sapply(list(bikes_C, bikes_F, bikes_Y), AIC),
               BIC = sapply(list(bikes_C, bikes_F, bikes_Y), BIC)) |>
  knitr::kable(digits = c(0, 3, 1, 1))
```

Model	adj_R2	AIC	BIC
C	0.483	8459.3	8488.8
C + season	0.554	8387.8	8429.9
C + year	0.744	8108.9	8142.6

```
confint(bikes_F)[c("weekendTRUE", "holidayyes"), ]
```

```
                2.5 %    97.5 %
weekendTRUE  -995.0281 -580.2357
holidayyes  -1705.2946 -482.9267
```

```
confint(bikes_Y)[c("weekendTRUE", "holidayyes"), ]
```

```
                 2.5 %    97.5 %
weekendTRUE  -955.0442 -640.9895
holidayyes  -1556.0113 -630.0647
```

Both comparisons are nested (C is contained in each larger model). Both additions are strongly supported by every tool:

- **Adding `season`** ($d = 3$): $F = 27.5$, p < 0.001; adjusted R^2 rises from 0.483 to 0.554; AIC and BIC both drop by more than 55.
- **Adding `year`** ($d = 1$): $F = 504.5$, p < 0.001; adjusted R^2 jumps to 0.744; AIC and BIC both drop by more than 340. Ridership grew a lot from 2011 to 2012, and the index plot of residuals for model C shows this trend over time.

Our conclusions about weekends and holidays do not change much: the estimates are nearly the same (about 798 fewer rides on weekends and 1093 fewer on holidays with `year` in the model), but the confidence intervals get narrower when we explain more of the variability (the weekend interval with `year` is −955 to −641). Since the question is about the weekend and holiday effects "after accounting for" other things, a model with `year` (and possibly `season`) is arguably the better final model.

Function quick reference

Function	Purpose
`lm()`	Fit a linear model
`cor()`	Correlation matrix for a set of quantitative variables
`anova(reduced, full)`	Extra sums of squares F test comparing two nested models
`summary(model)$adj.r.squared`	Adjusted R^2 for a fitted model
`AIC()`, `BIC()`	Information criteria for a fitted model (smaller is better)
`confint()`	Confidence intervals for the coefficients of a fitted model